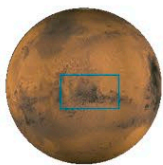


TERRA MERIDIANI

Meridiani Planum



TYPE Highland plain
AGE Over 3.5 billion years
DIAMETER 680 miles (1,100 km)

In the westernmost portion of Terra Meridiani and just south of the equator lies the high plain Meridiani Planum. It does not stand out in the

global view of Mars but achieved prominence as the landing site and exploration ground for the Opportunity rover. The plain is about 15° due west of Schiaparelli

Crater (see p.164). Smaller impact craters pepper the area. They range from Airy, just 26 miles (41 km) across, to much smaller bowl-shaped craters, such as the 72-ft (22-m) wide Eagle Crater where Opportunity landed.

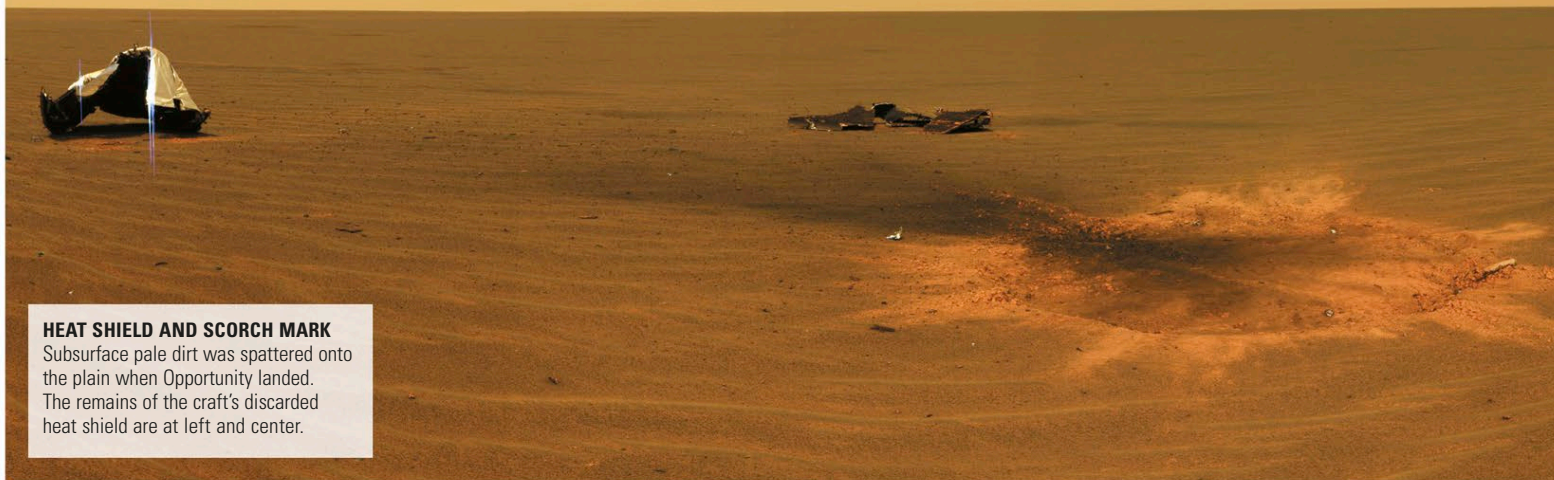


UNIQUE METEORITE

This basketball-sized rock has an iron-nickel composition. It is not a Martian rock but a meteorite—the first to be found on a planet other than Earth.

Volcanic basalt is found within the area, but the region is of the greatest interest because it contains ancient layered sedimentary rock that includes the mineral hematite. Some of this mineral, which on Earth almost always

forms in liquid water, is exposed and easily found on the surface. The hematite could have been produced from iron-rich lavas, but it is believed that water was involved. This area is dry now but was once soaking wet and could well have been the site of an ancient lake or sea about 3.7 billion years ago. Eroded layered outcrops beyond the landing site support this theory and point to a deep and long-lasting volume of water as large as Earth's Baltic Sea. At this time, Mars must have been a much warmer and wetter place than it is today.



HEAT SHIELD AND SCORCH MARK

Subsurface pale dirt was spattered onto the plain when Opportunity landed. The remains of the craft's discarded heat shield are at left and center.

PROMETHEI TERRA

Reull Vallis



TYPE Outflow channel
AGE 2–3.5 billion years
LENGTH 587 miles (945 km)

Reull Vallis is one of the larger channels of the southern hemisphere. It extends across the northern part of Promethei Terra, to the east of Hellas Basin (see p.165). Reull is thought to have had a complex evolution because it exhibits the characteristics of all

three channel types seen on Mars. In the collapsed region at the southern base of the volcano Hadriaca Patera, for example, it is a fully formed outflow channel. But small tributaries also feed into the main channel, as they would in a runoff channel. And the main channel has the features of a fretted channel—a wide, flat floor and steep walls. Reull Vallis takes its name from the Gaelic word for “planet.”

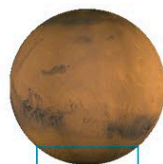
MERGING CHANNELS

Reull Vallis (top left) is joined by a tributary, Teviot Vallis (right). The parallel structures in the fretted channel floor were possibly caused by a glacial flow of loose debris mixed with ice.



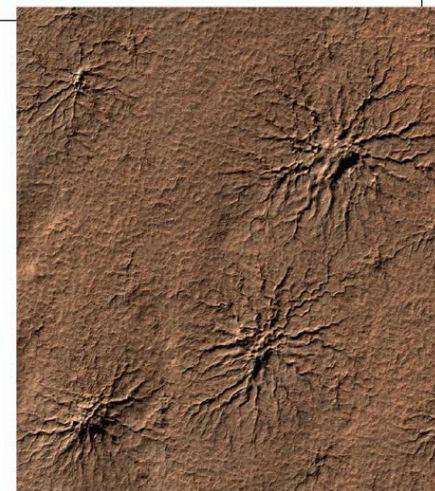
PLANUM AUSTRALE

South Polar Region



TYPE Polar ice cap
AGE Under 2.5 billion years
DIAMETER 900 miles (1,450 km)

The South Polar Cap, known formally as Planum Australe (Southern Plain), is an ice-dominated mound a few miles high. It consists of three different parts. First is the bright polar cap that is roughly centered on the South Pole. This is a permanent cap of water ice with a covering of carbon-dioxide ice. Next are the scarps, made primarily of water ice, which fall away from the cap to the surrounding plains. Third, hundreds of square miles of permafrost encircle



SPIDERS ON MARS

These spiderlike features in the south polar region of Mars were cut by dry ice (frozen carbon dioxide) as it turned to gas in the spring. The channels are 3–6 ft (1–2 m) deep.

the region. The permafrost is water ice mixed into the soil and frozen to the hardness of solid rock. The South Polar Cap shrinks and grows with the seasons like the North Polar Cap (see p.161). Yet surprisingly the southern cap does not get warm enough in the summer to lose its carbon-dioxide ice covering. Dust storms that block out the Sun may keep the cap cooler than expected.

SOUTH POLAR CAP

Carbon-dioxide frost (shown as pink) covers over the water-ice cap (green-blue). Scarps of water ice at the edge of the cap slope toward the surrounding plains.

